



ABES ENGINEERING COLLEGE, GHAZIABAD (AUTONOMOUS)

Make-Up Test (Session: 2025-26)

Roll No.:

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Course Name: B.Tech
Subject Name: Data Structure
Subject Code: 25CS101

Max. Marks: 15
Time: 75 Minutes
Semester: I

Note:

1. Keeping Mobile Phone/Any Electronic Gadgets, Even in OFF position is treated as UFM
2. Don't Write Anything on Question Paper
3. Attempt All Questions
4. Assume Any Missing Data if required

SECTION-A (Maximum Knowledge Level: K2)				
Q. No.	Question	CO	KL	PI
1. Attempt All Questions		(1 x 6)		
(a)	Why are loader and linker important in program execution?	CO1	K1	2.2.1
(b)	Explain source code, object code, and executable code with a simple diagram.	CO1	K2	1.2.1
(c)	In a railway booking system, which data type would you use to store: 1. Passenger age 2. Ticket amount (₹1200.50)	CO2	K2	1.3.5
(d)	Define identifier. Write four valid and two invalid identifier examples.	CO2	K2	1.4.1
(e)	How do you declare and initialize a 1D array in C++?	CO3	K2	1.2.1
(f)	What is the output of the following C++ Program? <pre>#include<iostream.h> int main() { int ary[] = {10,20,30}, for (int i=0;i<3;i++) { cout<< ary[i]+1<<" "; } }</pre>	CO3	K2	1.2.1
SECTION-B				
2. Attempt Any one Question		(Same K Levels) (3 x 1)		
(a)	Explain different types of Tokens in C++ with examples.	CO1	K2	2.1.1
(b)	Explain the basic structure of a C++ program with an example.	CO1	K2	2.1.2
SECTION-C				
3. Attempt Any one Question		(Same K Levels) (3 x 1)		
(a)	Write a C++ program to input a number and check whether it is even or odd using ternary operator.	CO2	K3	1.2.3
(b)	Write a C++ program to input the marks of three subjects and calculate the average marks. Display the result.	CO2	K3	1.2.3

SECTION-D				
4. Attempt Any one Question		(Same K Levels) (3 x 1)		
(a)	Write a program in C++ to find the sum of all elements of a 2D matrix and take input from user.	CO3	K3	2.2.3
(b)	A company gives insurance to its drivers in the following cases: (i) If the driver is married. (ii) If the driver is unmarried, male & above 30 years of age. (iii) If the driver is unmarried, female & above 25 years of age. In all other cases, the driver is not insured. If the marital status, gender and age of a driver are inputs, write a program to determine whether the driver is to be insured or not.	CO3	K3	2.2.1

CO Course Outcomes mapped with respective question

KL Bloom's knowledge Level (K1, K2, K3, K4, K5, K6)

K1-Remember, K2-Understand, K3-Apply, K4-Analyze, K5-Evaluate, K6-Create